

Dataset:

Height (m)	Favourite Color	Gender	Weight (kg)
1.6	Blue	Male	88
1.6	Green	Female	76
1.5	Blue	Female	56
1.8	Red	Male	73
1.5	Green	Male	77
1.4	Blue	Female	57

Iteration 1

$$\text{Residual} = \text{Actual target} - \text{Predicted target} = 88 - 71.2$$

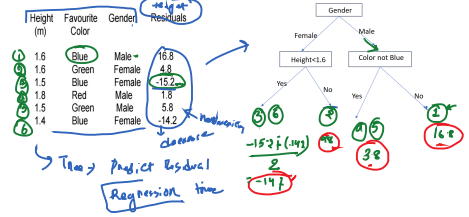
Step1: Calculate (Initial) Prediction

Height (m)	Favourite Color	Gender	Weight (kg)
1.6	Blue	Male	88
1.6	Green	Female	76
1.5	Blue	Female	56
1.8	Red	Male	73
1.5	Green	Male	77
1.4	Blue	Female	57

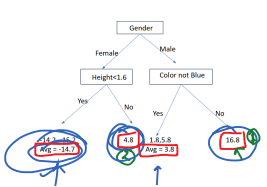
Step2: Calculate Residual

Height (m)	Favourite Color	Gender	Weight (kg)	(Initial) Prediction	Residual
1.6	Blue	Male	88	71.2	16.8
1.6	Green	Female	76	71.2	4.8
1.5	Blue	Female	56	71.2	-15.2
1.8	Red	Male	73	71.2	1.8
1.5	Green	Male	77	71.2	5.8
1.4	Blue	Female	57	71.2	-14.2

Step3: Build a tree to predict the residuals



Tree after 1st Iteration



Iteration 2

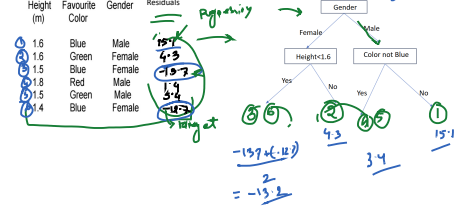
Step1: Calculate (updated) Prediction

Height (m)	Favourite Color	Gender	Weight (kg)	Updated Prediction
1.6	Blue	Male	88	71.2
1.6	Green	Female	76	71.2
1.5	Blue	Female	56	71.2
1.8	Red	Male	73	71.2
1.5	Green	Male	77	71.2
1.4	Blue	Female	57	71.2

Step2: Calculate Residual

Height (m)	Favourite Color	Gender	Weight (kg)	(Updated) Prediction	Residual
1.6	Blue	Male	88	71.2	16.8
1.6	Green	Female	76	71.2	4.8
1.5	Blue	Female	56	71.2	-15.2
1.8	Red	Male	73	71.2	1.8
1.5	Green	Male	77	71.2	5.8
1.4	Blue	Female	57	71.2	-14.2

Step3: Build a tree to predict the residuals



(after 1st iteration Updated) prediction = Initial prediction + L.R. x prediction (of Residuals)

$$\text{2nd example} = 71.2 + 0.1 \times 4.8 = 71.68$$

Iteration 3

Step1: Calculate (updated) Prediction

Height (m)	Favourite Color	Gender	Weight (kg)	(Updated) Prediction
1.6	Blue	Male	88	71.2
1.6	Green	Female	76	71.2
1.5	Blue	Female	56	71.2
1.8	Red	Male	73	71.2
1.5	Green	Male	77	71.2
1.4	Blue	Female	57	71.2

Step2: Calculate Residual

Height (m)	Favourite Color	Gender	Weight (kg)	(Updated) Prediction	Residual
1.6	Blue	Male	88	71.2	16.8
1.6	Green	Female	76	71.2	4.8
1.5	Blue	Female	56	71.2	-15.2
1.8	Red	Male	73	71.2	1.8
1.5	Green	Male	77	71.2	5.8
1.4	Blue	Female	57	71.2	-14.2

Step3: Build a tree to predict the residuals

Height (m)	Favourite Color	Gender	Residuals
1.6	Blue	Male	16.8
1.6	Green	Female	4.8
1.5	Blue	Female	-15.2
1.8	Red	Male	1.8
1.5	Green	Male	5.8
1.4	Blue	Female	-14.2

Now build a 3-regression tree to predict the latest residuals.
Now prediction of an example = initial guess + 0.1 x prediction by 1-tree + 0.1 x prediction by 2-tree + 0.1 x prediction by 3-tree + ... until the maximum number of iterations is done or the updated residuals become insignificant

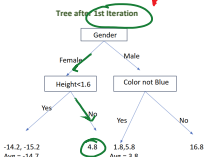
Stopping Criteria

- Maximum number of tree
- Until the residuals become insignificant

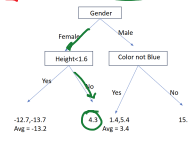
(after 2nd iteration Updated) prediction = Initial prediction + L.R. x prediction (of Residuals)

$$= 71.68 + 0.1 \times 4.3 = 71.99$$

Prediction from Trained model



Tree after 2nd Iteration



New data

Height (m)	Favourite Color	Gender	Weight (kg)
1.7	Green	Female	72.1

Predicted initial prediction

$$= 72.1 + 0.1 \times 4.3 + 0.1 \times 4.3 = 72.11$$

